

Final Review

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Exercise 1.

- a) Do problem 4.59 from Griffiths: If two electrons are in the spin singlet state, $S_a^{(1)}$ is the component of spin angular momentum of particle 1 along the axis defined by a vector $\mathbf{a}$, and $S_b^{(2)}$ is the spin angular momentum of particle 2 along the axis defined by some other vector $\mathbf{b}$, show that

$$\langle S_a^{(1)} S_b^{(2)} \rangle = -\frac{\hbar^2}{4} \hat{a} \cdot \hat{b}.$$

- b) Redo problem 4.59 from Griffiths by using Exercise 4(c) from the midterm review: If two electrons are in the spin singlet state, $S_z^{(1)}$ is the component of spin angular momentum of particle 1 along the z -axis, and $S_\theta^{(2)}$ is the spin angular momentum of particle 2 along the $\hat{r} = (\theta, 0)$ axis, show that

$$\langle S_z^{(1)} S_\theta^{(2)} \rangle = -\frac{\hbar^2}{4} \cos \theta.$$

- a) We can assume that $\hat{a} = \hat{z}$ and $\hat{b}$ is in the (x, z) -plane. Thus, $S_a^{(1)} = S_z^{(1)}$ and $S_b^{(2)} = \cos(\theta) S_z^{(2)} + \sin(\theta) S_x^{(2)}$. We can then compute

$$\frac{1}{\sqrt{2}} \langle \text{singlet} | S_a^{(1)} S_b^{(2)} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) = -\frac{1}{\sqrt{2}} \langle \text{singlet} | \left(\frac{\hbar^2}{4} \cos \theta |\uparrow\downarrow\rangle + \frac{\hbar^2}{4} \cos \theta |\downarrow\uparrow\rangle \right),$$

since we can ignore the terms with $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ which will get killed by taking the bracket with the singlet state. Thus, we can compute this bracket to be

$$-\frac{1}{2} \cdot \frac{\hbar^2}{4} \cdot 2 \cos \theta = -\frac{\hbar^2}{4} \cos \theta.$$

- b) By Exercise 4(c) from the Midterm 2 Review, we know that the rotation by angle θ about the y -axis can be written

$$\begin{aligned} U_y(\theta) &= e^{-\frac{i}{2}\theta\sigma_y} \\ &= \cos(\theta/2) - i\sigma_y \sin(\theta/2). \end{aligned}$$

Now, we need to determine S_θ . To do this, we need to find the matrix whose eigenvectors are $U_y(\theta)|\uparrow\rangle$ and $U_y(\theta)|\downarrow\rangle$. At this point, we can use Exercise 4 of Midterm 2 (the midterm, not the review). In this exercise, we learned that if H is a matrix with eigenvector $|\psi\rangle$ and U is a unitary operator, then $UHU^\dagger$ has eigenvector $U|\psi\rangle$ with the same eigenvalue. Thus, S_θ will be given by $U_y(\theta)S_zU_y(\theta)^\dagger$. We can compute this, using Exercise 4(c) from the Midterm 2 Review, as

$$\begin{aligned}\sigma_\theta &= U_y(\theta)\sigma_zU_y(\theta)^\dagger \\ &= \sigma_z \cos^2(\theta/2) + i[\sigma_z, \sigma_y] \sin(\theta/2) \cos(\theta/2) + \sigma_y \sigma_z \sigma_y \sin^2(\theta/2) \\ &= \sigma_z [\cos^2(\theta/2) - \sin^2(\theta/2)] + 2\sigma_x \sin(\theta/2) \cos(\theta/2) \\ &= \sigma_z \cos \theta + \sigma_x \sin \theta \\ &= \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}.\end{aligned}$$

Note that $S_\theta = \frac{\hbar}{2}\sigma_\theta$. At this point, it's easiest to figure out what to do if we write the spin singlet state in vector notation:

$$\frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right).$$

Thus, we get

$$\begin{aligned}\langle S_z^{(1)} S_\theta^{(2)} \rangle &= \frac{\hbar^2}{8} \left(\begin{bmatrix} 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \end{bmatrix} \right) \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} \sin \theta \\ -\cos \theta \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \right) \\ &= \frac{\hbar^2}{8} (-\cos \theta - \cos \theta) \\ &= -\frac{\hbar^2}{4} \cos \theta.\end{aligned}$$

Exercise 2. Hidden Symmetries. In classical mechanics, $\frac{1}{r}$ potentials have an additional conserved quantity that is rarely covered in introductory courses. This quantity is called the **Runge-Lenz vector**, and it is given by

$$\mathbf{F} = \frac{1}{m} \mathbf{p} \times \mathbf{L} - \frac{\gamma}{r} \mathbf{r},$$

where γ is the constant associated to the potential $V(r) = -\gamma/r$, e.g. $\gamma = e^2/4\pi\epsilon_0$ or $\gamma = MG$.

- a) If we replace all the classical dynamical variables in the above expression by quantum operators, explain why the result is ambiguous.

Hints: When we upgrade $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ to a quantum operator, note that $\mathbf{L} = -\mathbf{p} \times \mathbf{r}$ as operators. Why? Does $\mathbf{p} \times \mathbf{L} = -\mathbf{L} \times \mathbf{p}$ as operators?

- b) It turns out¹ that the correct quantum mechanical version of $\mathbf{F}$ is

$$\mathbf{F} = \frac{1}{2m} (\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p}) - \frac{\gamma}{r} \mathbf{r}.$$

Show that for the hydrogen atom hamiltonian $H = p^2/2m - \gamma/r$, $[H, \mathbf{F}] = \mathbf{0}$, so that $\mathbf{F}$ is a symmetry.

- a) The result is ambiguous because $\mathbf{p}$ and $\mathbf{L}$ don't commute. We can check this directly:

$$[p_i, r_j p_k] \varepsilon_{jkl} = [p_i, r_j] p_k \varepsilon_{jkl} = -i\hbar \delta_{ij} p_k \varepsilon_{jkl} = p_k \varepsilon_{ikl} \neq 0.$$

This implies that $\mathbf{p} \times \mathbf{L} \neq -\mathbf{L} \times \mathbf{p}$; indeed, to obtain an equality, we would need to add additional terms proportional to the commutator computed above. Thus, we can't just upgrade $\mathbf{F}$ to a quantum operator without first figuring out the right linear combination of $\mathbf{p} \times \mathbf{L}$ and $\mathbf{L} \times \mathbf{p}$ to take.

- b) In order to compute the commutator $[H, \mathbf{F}]$, there will be four terms which we need to compute, since H and $\mathbf{F}$ are each the sum of two terms. However, we can immediately argue that the terms

$$[p^2, (\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p})] = [V(r), \mathbf{r}/r] = 0.$$

For the second one, this is obvious because r commutes with all components of $\mathbf{r}$ and with itself. For the first one, we can compute

$$[p^2, (\mathbf{p} \times \mathbf{L})_k] = [p^2, p_i L_j \varepsilon_{ijk}] = p_i [p^2, L_j] \varepsilon_{ijk},$$

since p^2 commutes with all components of $\mathbf{p}$. Now, p^2 is a scalar with respect to rotations; in other words, rotating the coordinate system does not change the length of vectors. On the other hand, $\mathbf{L}$ is orbital angular momentum which precisely generates rotations of 3-dimensional space in quantum mechanics. It follows that p^2 and $\mathbf{L}$ must commute. We're left with the cross terms, and we work one at a time.

$$[p^2, r_i/r] = p_j [p_j, r_i/r] + [p_j, r_i/r] p_j = -i\hbar \left[p_j \left(\frac{\delta_{ij}}{r} - \frac{r_i r_j}{r^3} \right) + \left(\frac{\delta_{ij}}{r} - \frac{r_i r_j}{r^3} \right) p_j \right].$$

Now, we can compute

$$[p_j, 1/r] = i\hbar \frac{r_j}{r^3},$$

and

$$[p_j, r_i r_j / r^3] = -i\hbar \left(\frac{2r_i}{r^3} - \frac{3r_i r_j r_j}{r^5} \right) = i\hbar \frac{r_i}{r^3},$$

¹The way you would prove this is by matching Poisson bracket relations with $\mathbf{F}$ in classical mechanics to corresponding ones in quantum mechanics (upgrading the Poisson brackets to commutators). This would then allow you to determine the right combination of $\mathbf{p} \times \mathbf{L}$ and $\mathbf{L} \times \mathbf{p}$ to take.

since $r_j r_j = r^2$. Thus, we can move all the p_j to the right in the expression for $[p^2, r_i/r]$ to obtain

$$[p^2, r_i/r] = -i\hbar \left(\frac{\delta_{ij}}{r} - \frac{r_i r_j}{r^3} \right) p_j - \hbar^2 \frac{2r_i}{r^3}.$$

Now, we can work with the other cross term. Its k -component is given by

$$\begin{aligned} [1/r, (\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p})_k] &= \left(\left[\frac{1}{r}, p_i \right] L_j - L_i \left[\frac{1}{r}, p_j \right] \right) \varepsilon_{ijk} \\ &= -i\hbar \left(\frac{r_i}{r^3} L_j - L_i \frac{r_j}{r^3} \right) \varepsilon_{ijk}, \end{aligned}$$

where you should note that $1/r^3$ commutes with $\mathbf{L}$ since it's also a scalar with respect to rotations. Now,

$$[r_i, L_j] = [r_i, r_l p_m] \varepsilon_{lmj} = r_l [r_i, p_m] \varepsilon_{lmj} = i\hbar r_l \delta_{im} \varepsilon_{lmj} = i\hbar r_l \varepsilon_{lij}.$$

Since $\varepsilon_{lij} \varepsilon_{ijk} = \varepsilon_{ijl} \varepsilon_{ijk} = 2\delta_{kl}$, we can move the r_i to the right in our expression for $[1/r, (\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p})_k]$ to obtain

$$[1/r, (\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p})_k] = -\frac{i\hbar}{r^3} \varepsilon_{ijk} (r_i L_j + L_j r_i) = -\frac{2i\hbar}{r^3} (\mathbf{r} \times \mathbf{L})_k + 2\hbar^2 \frac{r_k}{r^3}.$$

We can now use the triple product identity “BAC-CAB:” Since $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, we have

$$\mathbf{r} \times \mathbf{L} = \mathbf{r} \times (\mathbf{r} \times \mathbf{p}) = \mathbf{r}(\mathbf{r} \cdot \mathbf{p}) - r^2 \mathbf{p},$$

where you need to be careful to get the order right (the “CAB” at the end is actually “ABC”), since that's how the operators appear in $\mathbf{r} \times \mathbf{L}$. Thus, switching the index $i \leftrightarrow k$, we get

$$[1/r, (\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p})_i] = -2i\hbar \left(\frac{r_i r_j}{r^3} p_j - \frac{1}{r} p_i \right) + 2\hbar^2 \frac{r_i}{r^3}.$$

Noting that $\frac{1}{r} \delta_{ij} p_j = \frac{1}{r} p_i$ and adding in the factors of γ and $\frac{1}{2m}$, we see that this commutator precisely cancels the previous commutator with p^2 .

Exercise 3.

- Show that the average momentum vanishes, $\langle \mathbf{p} \rangle = \mathbf{0}$, in any bound state of the hamiltonian $H = p^2/2m + V(\mathbf{x})$.
- The following alleged solution to (a) is given in a certain textbook.

Since $p_x = \frac{im}{\hbar} [H, x]$, it follows that

$$\langle p_x \rangle = \frac{im}{\hbar} (\langle \psi | H x | \psi \rangle - \langle \psi | x H | \psi \rangle).$$

Using $H |\psi\rangle = E |\psi\rangle$ and $\langle \psi | H = \langle \psi | E$, we obtain $\langle p_x \rangle = 0$, hence $\langle \mathbf{p} \rangle = \mathbf{0}$.

This argument, if valid, would imply that $\langle \mathbf{p} \rangle = \mathbf{0}$ in *any* stationary state, bound and unbound. But the counterexample $V(\mathbf{x}) \equiv 0$, $\psi(\mathbf{x}) = \exp(i\mathbf{k} \cdot \mathbf{x})$ shows that there is a mistake in the argument. What is it?

The argument given in (b) does prove (a), since for bound states ψ is normalizable. For an unbound state with non-normalizable ψ as given in (b), the bracket $\langle \psi | \mathbf{x} | \psi \rangle$ is not defined, so the calculation in (b) isn't valid.

Exercise 4. For the ground state of the hydrogen atom, calculate the probability that the electron and the proton are farther apart than would be permitted by classical mechanics at the same total energy.

We first need to figure out the classical calculation. Denote the energy of the system by E . We know that

$$E = \frac{1}{2}mv^2 - \frac{e^2}{r}.$$

On the other hand, assuming the electron is in a classical circular orbit, we must have

$$\frac{mv^2}{r} = \frac{e^2}{r^2},$$

so that

$$mv^2 = \frac{e^2}{r}.$$

Thus,

$$E = -\frac{e^2}{2r},$$

or

$$r_0 = -\frac{e^2}{2E}.$$

But note that this r_0 is precisely the Bohr radius a ! Then, the probability in question is

$$\frac{4}{a^3} \int_a^\infty r^2 e^{-2r/a} dr,$$

where we already evaluated the angular integrals. This can be computed by repeated integration by parts:

$$\begin{aligned} \frac{4}{a^3} \int_a^\infty r^2 e^{-2r/a} dr &= \frac{4}{a^3} \left(-\frac{a}{2} r^2 e^{-2r/a} \Big|_a^\infty + \frac{a}{2} \int_a^\infty r e^{-2r/a} dr \right) \\ &= \frac{4}{a^3} \left(\frac{a^3}{2} e^{-2} + \frac{a^3}{4} e^{-2} + \frac{a^2}{4} \int dr e^{-2r/a} \right) \\ &= \frac{7}{2} e^{-2} \approx 47\%. \end{aligned}$$

So the likelihood that the electron is farther away than classically permitted is nearly 50%!

Exercise 5. Bonus—This exercise is beyond the scope of the final.

- a) Consider a singly-ionized helium ion. How much more energy does it take to ionize its bound electron compared to hydrogen?
- b) Still with He^+ . What is the wavelength of the emitted photon during the electron transition from $n = 2 \rightarrow 1$?
- c) Now, consider the usual helium-4. Which ground state has higher energy, parahelium (spin singlet) or orthohelium (spin triplet)? Why? **Griffiths 5.14**. How would this change if the two electrons are identical bosons?
- d) Helium-3 is a fermion with spin-1/2 as compared to helium-4, which is a boson. Why?
- a) The only difference between singly-ionized helium and hydrogen is the number of protons. The energy levels for hydrogen-like atoms come from a potential in the Schrödinger equation which is proportional to Ze^2 . Now, we have to remember that in hydrogen the energy levels go as Z^2 . Explicitly,

$$E_n = -\frac{1}{2} \frac{\alpha^2 Z^2 m c^2}{n^2}$$

$$= -E_0 \frac{Z^2}{n^2},$$

where $E_0 = 13.6 \text{ eV}$. Thus, helium-3 is 4 times harder to ionize than hydrogen.

- b) Recall that $E = pc$ for light; hence,

$$E = \frac{hc}{\lambda}.$$

Since $E_2 - E_1 = 3E_0$, we have

$$\lambda = \frac{hc}{3E_0}.$$

Since $hc = 1240 \text{ eV} \cdot \text{nm}$, we have

$$\lambda \approx 30 \text{ nm}.$$

- c) Since the triplet is symmetric, the spatial wavefunction which is associated to orthohelium must be antisymmetric. Since the lowest energy state is symmetric (both electrons in $n = 1$), it follows that the triplet has higher energy than the singlet. If they were both bosons, then we'd have the opposite.
- d) Helium-3 is a fermion because it has an odd number of fermions; on the other hand, helium-4 is a boson because it has an even number of fermions.